

SOFTWARE REQUIREMENTS SPECIFICATION (SRS)

Project Name: CIPD 360 – Academic ERP System

This is initial requirement which we have prepared for scheduling, attendance, feedback, lecture uploads and dashboard

1. INTRODUCTION

1.1 Purpose

This document specifies the functional and non-functional requirements for the CIPD 360 ERP System, a web-based academic ERP platform intended to manage class scheduling, attendance, feedback, content delivery, reporting, and faculty honorarium calculations. The SRS consolidates requirements discussed during stakeholder meetings and expands them into implementable system specifications.

1.2 Scope

CIPD 360 is a role-based ERP system supporting Admins, Faculty, and Students. The initial release (MVP) focuses on:

- Class scheduling
- Attendance tracking using Wi-Fi router detection
- Per-session student feedback
- Content sharing
- Reports and dashboards

Future phases may extend to grading, assignments, gamification, and deeper analytics.

1.3 Definitions & Acronyms

Term	Meaning
ERP	Enterprise Resource Planning
BSSID	Basic Service Set Identifier (Wi-Fi router identifier)
Session	A single scheduled class occurrence
MVP	Minimum Viable Product
PII	Personally Identifiable Information

2. OVERALL DESCRIPTION

2.1 Product Perspective

CIPD 360 is a standalone web-based ERP accessible via browser. It may later integrate with CIPD's public website but will initially function as an independent portal.

2.2 User Classes and Characteristics

User	Description
Admin	Full system control, scheduling, attendance correction, reporting
Faculty	View schedules, upload content, view attendance & feedback
Student	View schedule, attend sessions, submit feedback, view content

2.3 Operating Environment

- Frontend: Web browser (desktop & mobile)
- Backend: Node.js (REST APIs)
- Database: PostgreSQL
- Hosting: Cloud-based (AWS / DigitalOcean / equivalent)

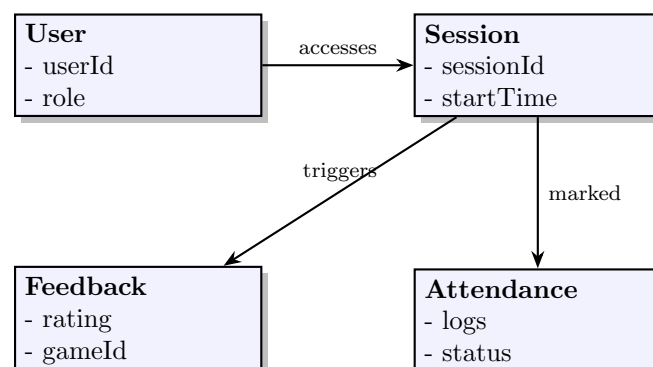
2.4 Design Constraints

- Attendance must primarily rely on campus Wi-Fi router detection
- Feedback must be short and high-completion
- System must support manual overrides by Admin
- Data privacy compliance is mandatory

3. SYSTEM ARCHITECTURE & UML

System Workflow Overview

UML Class Representation



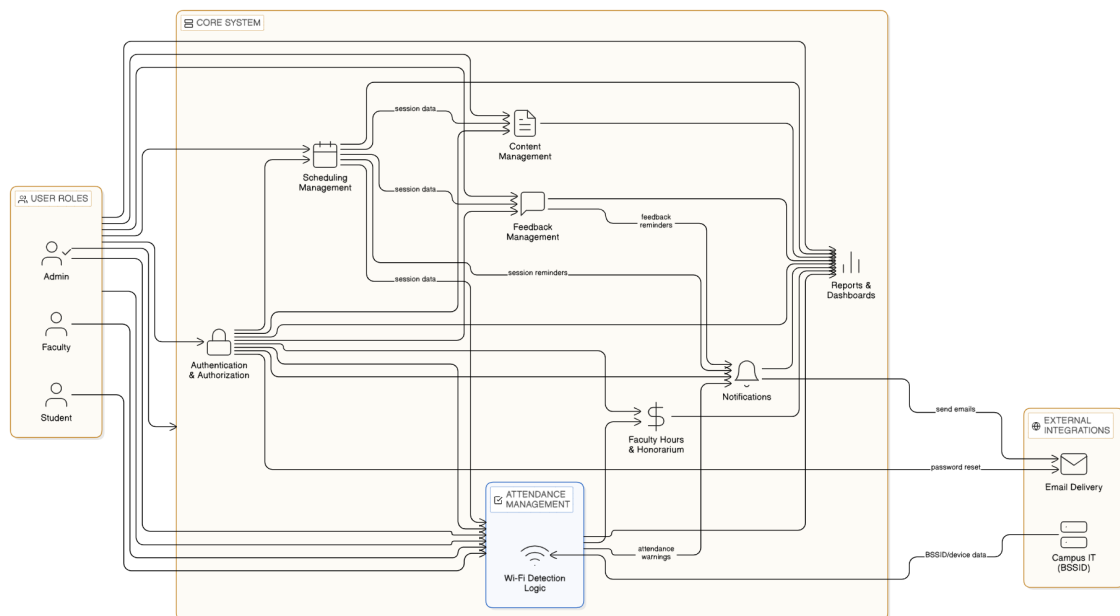
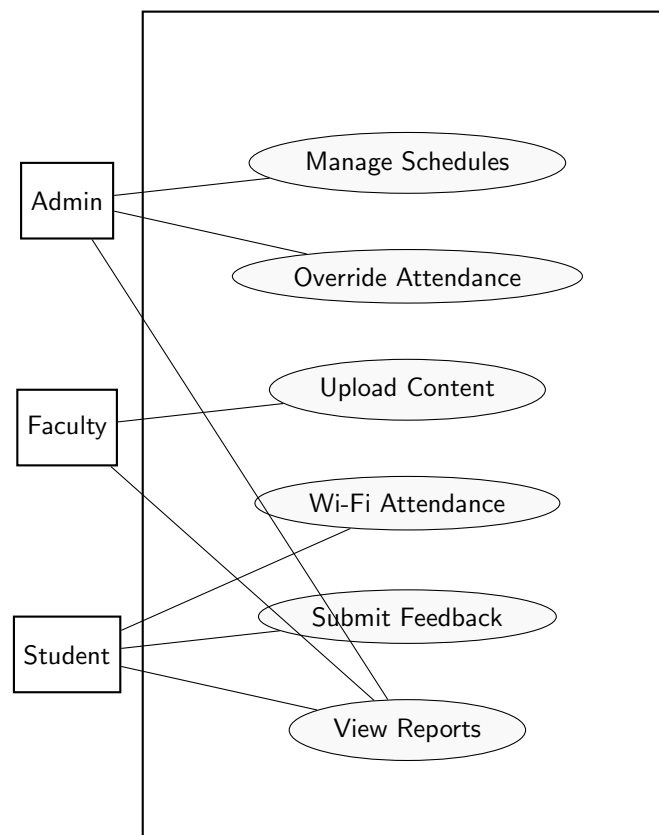
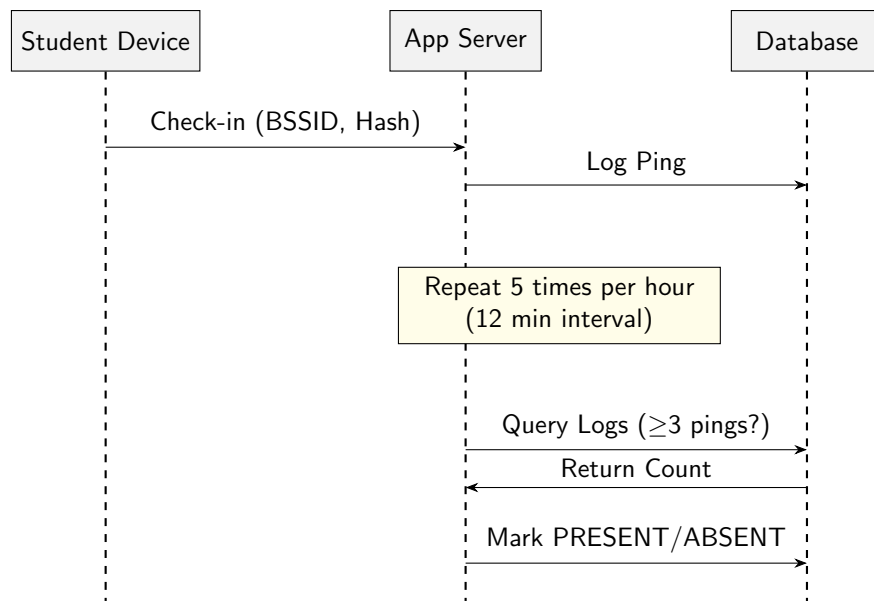


Figure 1: Academic ERP System Architecture

Use Case Diagram



Sequence Diagram: Wi-Fi Attendance Logic



3. SYSTEM FEATURES & FUNCTIONAL REQUIREMENTS

3.1 Authentication & Authorization

3.1.1 Description

Users must authenticate securely and be routed to role-specific dashboards.

3.1.2 Functional Requirements

- Email + password login
- Role-based access control (RBAC)
- Password reset via email
- JWT-based session handling
- Secure logout and session expiry

3.2 Scheduling Management

3.2.1 Description

Admins manage class schedules; students and faculty view personalized schedules.

3.2.2 Functional Requirements

- Admin can:
 - Create, edit, cancel sessions
 - Schedule multiple sessions per day (default up to 6, extendable)

- Assign faculty, venue, time, and date
 - Create recurring schedules (daily/weekly)
 - System must:
 - Detect scheduling conflicts (faculty, venue, student)
 - Allow override with admin justification
 - Auto-populate schedules on dashboards
 - Send email reminders before sessions
 - Students & faculty can:
 - View schedules in calendar and list views
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3.3 Attendance Management

3.3.1 Description

Attendance is recorded automatically using campus Wi-Fi connectivity based on access point.

3.3.2 Attendance Logic

- Each session has an attendance window
- System performs 5 pings each at constant interval (12 minutes) each hour and continues for entire time as given in schedule
- If a student device is detected ≥ 3 times per hour, the student is marked PRESENT
- Detection is based on BSSID matching for the venue which we will get from IT department

3.3.3 Functional Requirements

- System must:
 - Accept attendance ping logs (timestamp, device hash, BSSID)
 - Aggregate ping results per session
 - Automatically mark attendance
 - Store raw logs for audit
 - Admin can: Manually edit attendance; View attendance correction history
 - Students can: View their attendance percentage; View other attendance related analytics
 - Reports must be exportable (CSV/PDF)
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3.4 Feedback Management

3.4.1 Description

Students provide feedback after each session.

3.4.2 Functional Requirements

- Feedback must be collected per session
- Question types: 1–5 rating scale, Yes/No questions, Conditional follow-ups

- Categories: Structure, Content, Logistics & Support
 - Feedback must be completable within 60 seconds
 - Admin can: Configure mandatory feedback; Trigger reminders; Generate analytics
 - Faculty can: View aggregated feedback (no student identity)
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3.5 Gamified Feedback

3.5.1 Description

Feedback may be collected via a simple game to increase engagement.

3.5.2 Requirements

- Initial game: simple (e.g., Snake); Game duration: 30–60 seconds
 - Feedback embedded within gameplay; Game must not block non-game fallback
 - Future: 2-3 more games used cyclically; Each game used to collect feedback
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3.6 Content Management

3.6.1 Description

Faculty share session-related learning materials.

3.6.2 Functional Requirements: Upload files (PPT, PDF, links)

Content linked to sessions

Role-based permissions

Secure download

Timestamped uploads.

3.7 Faculty Hours & Honorarium

3.7.1 Description

Faculty payment calculations are derived from session data.

3.7.2 Functional Requirements: System calculates sessions conducted and total hours.

Admin can adjust hours manually and generate monthly honorarium reports in PDF/CSV.

3.8 Notifications

3.8.1 Description

System sends automated notifications.

3.8.2 Functional Requirements: Email and in-app alerts for:

reminders

schedule changes

attendance warnings

Support for Calendar invites (ICS).

3.9 Reports & Dashboards

3.9.1 Description

Visual summaries for all stakeholders.

3.9.2 Reports:

Student attendance trends
Session-wise feedback analysis
Faculty workload summaries
Admin overview dashboards

4. NON-FUNCTIONAL TECHNICAL REQUIREMENTS

(Future work - not mandatory but desirable as of now)

4.1 Performance: API response < 500ms; scalable attendance processing.

4.2 Security: HTTPS only; Secure password hashing; Role-based API protection; Audit logs.

4.3 Scalability: Stateless backend; Background job queues; Horizontal scaling support.

4.4 Reliability: Daily backups; Fault-tolerant design; Graceful degradation.

4.5 Compliance: Data privacy & consent handling; Configurable retention policies.

5. ASSUMPTIONS & DEPENDENCIES

- Campus IT will provide Wi-Fi router BSSID data
 - Students use personal devices connected to campus Wi-Fi (mobile-S)
 - Email delivery infrastructure is available
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6. PENDING DECISIONS QUESTIONS

1. Front end layout and UI.
 2. Hosting and domain ownership.
 3. Type of questions in feedback.
 4. Grade Management Portal.
 5. Final attendance data source format.
 6. Guest faculty honorarium rules.
 7. Language support.
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7. FUTURE ENHANCEMENTS

- Assignment and grading system
 - Inventory Management System
 - Advanced analytics and summary generation
 - Mobile app and multiple gamification options
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8. ACCEPTANCE CRITERIA (MVP)

- Role-based login operational; Scheduling with conflict detection
- Automated attendance marking; Feedback and Content uploads per session
- Admin reports available; Manual overrides audited